

I. A FRAME WORK OF A SMART THERAPEUTIC INSOLE SYSTEM BY MEASURING PLANTAR PRESSURE FOR EARLY DIAGNOSIS OF DIABETIC FOOT ULCERS USING MACHINE LEARNING AND THERMAL IMAGING TECHNIQUES

Diviya S
Department of Biomedical Engineering
(Student)
SRM Easwari Engineering College
(Anna University)
Chennai, India
diviyasankar08@gmail.com

Sangamitra N
Department of Biomedical Engineering
(Student)
SRM Easwari Engineering College
(Anna University)
Chennai, India
sangamitrnanagarajan@gmail.com

Venuhasini S M
Department of Biomedical Engineering
(Student)
SRM Easwari Engineering College
(Anna University)
Chennai, India
venuhasini514@gmail.com

Dr. B. Suresh Chander Kapali
Department of Biomedical Engineering
(Assistant Professor)
SRM Easwari Engineering College
(Anna University)
Chennai, India
sureshchander.b@eec.srmmp.edu.in

Dr. A. Saravan
Department of Computer Science &
Engineering (Professor & Vice Dean)
Sri Ramachandra Faculty of
Engineering and Technology
Chennai, India
vicedean@sret.edu.in

Abstract — Diabetic Foot Ulcers (DFUs) are a significant complication of diabetes, often resulting in severe morbidity, infections, and amputations. Early diagnosis of DFUs is critical in preventing these complications. This paper focuses on the frame work of a smart therapeutic insole system that integrates machine learning and thermal imaging techniques to measure plantar pressure and other foot parameters in diabetic patients. The system leverages pressure sensors, temperature sensors, and gait analysis to provide real-time data on plantar pressure distribution and potential risk areas. In this connection, many researchers have suggested a few techniques for early detection of DFUs. Their proposed works have been reviewed outlining the technological advancements in smart insole systems, their effectiveness in monitoring plantar pressure, and the application of machine learning for predictive analytics. The combination of machine learning models and thermal imaging has shown great potential in enhancing the accuracy of early DFU detection, improving patient outcomes, and reducing healthcare costs. This paper proposed a frame work and also discusses the different sensor types, methodologies, and design considerations for early diagnosis of diabetic foot ulcers using machine learning and thermal imaging techniques.

Keywords — Diabetic Foot Ulcers, International Diabetics Federation, Laser Induced Graphene, Infrared thermal imaging, Micro-Electromechanical Systems, Multi-Walled Carbon nanotubes, Polydimethylsiloxane, Internet of Medical Things.

Introduction

According to the IDF, 10.5% of adults worldwide aged between 20–79 years have diabetes. Nearly 15 percent of patients with diabetes are affected by Diabetic Foot

Ulcers. Diabetes mellitus is a chronic disease affecting millions worldwide, with diabetic foot ulcers (DFUs) being one of the most common complications. DFUs are caused by a combination of factors, including prolonged pressure on the foot, poor circulation, and neuropathy, which decreases the patient's ability to sense foot injuries. The consequence of untreated DFUs can be severe, leading to infections, gangrene, and ultimately, lower-limb amputations.

Recent advancements in wearable health technology have provided new solutions to prevent DFUs. Smart insole systems equipped with pressure and temperature sensors, as well as other physiological monitoring tools, offer a novel approach to the early detection and prevention of DFUs.

Machine learning (ML) and thermal imaging techniques have also emerged as powerful tools for analysing sensor data to identify patterns indicative of DFU risk. This paper explores the various designs and methodologies used in smart insole systems, their applications in monitoring diabetic foot health, and the role of machine learning and thermal imaging in enhancing the accuracy and timeliness of DFU detection.

RELATED WORKS

Stanley I.R. Okoduwa et al. focuses on the integration of custom-made therapeutic footwear with embedded sensors for monitoring plantar pressure and foot temperature in real time. By reducing plantar pressure through specialized insoles and offloading mechanisms, the smart footwear helps prevent the occurrence of DFUS. It demonstrated significant effectiveness in lowering forefoot pressure by up to 50%. The system is equipped with Bluetooth technology, enabling wireless data transmission to clinicians and providing real- time feedback to patients. The integration of sensors with a focus on patient- specific

needs is highlighted as a key innovation in this research work. [1]

Moulacia et al. developed a smart shoe with eight pressure sensors (FSR 402), two temperature and humidity sensors (SHT31), and an Arduino-based system to monitor foot conditions in diabetic patients. The smart shoe tracks pressure at four anatomical locations: the toe, first metatarsal head, fifth metatarsal head, and heel. The system provides real-time data via an Android app, with alerts issued when parameters exceed pre-set thresholds. The authors emphasized the system's sensitivity (100%) in pressure measurements, making it a reliable tool for DFU prevention. [2]

In the research work by Sarah Ostadabbas et al. introduces the Sparse-Sensing Continuous Plantar Pressure Model (SCPM) leverages Gaussian Mixture Models (GMMs) and Principal Component Analysis (PCA) to reconstruct high-resolution pressure maps using a limited set of sensors. The authors discuss the advantages of reducing sensor count while still maintaining an accurate representation of pressure distribution under the foot. By reducing dimensionality with PCA, the system efficiently monitors plantar pressure and provides early detection of high-risk areas for DFU development. [3]

Jones et al. have discussed the importance of accurately predicting diabetic foot ulceration (DFU) to reduce the burden of chronic wound management and amputation. Current models primarily focus on isolated factors such as temperature, pressure, or shear but fail to capture the complete foot microclimate. Their systematic review highlights the potential of using microclimate sensor arrays, combining variables like temperature, pressure, shear, and humidity, to develop a more effective real-time DFU prognostic model. They conclude that understanding these variables collectively may improve DFU prediction and emphasize the need for further research into multi-sensing systems. [4]

Lin et al. have presented a wearable smart insole with 48 textile-based pressure sensors to continuously monitor gait in real-time. The insole captures plantar pressure data and transmits it via Bluetooth to a smartphone app. The system's primary focus is on gait analysis, step count, and center of pressure shifts, which are crucial indicators for fall prevention and rehabilitation in patients with gait disorders. Its lightweight and unobtrusive design ensure comfort during extended use, making it suitable for daily monitoring. [5]

Li-Ying Zhang et al. have explored the use of Multilayer Perceptron (MLP) models for predicting plantar pressure distributions in diabetic patients based on ink footprint images. By analysing pressure across different foot regions (toes, metatarsal heads, midfoot, and heel), the study demonstrated how machine learning can enhance the prediction of pressure distribution improve recommendations for insole materials. [6]

Sousa et al. have discussed a three-step protocol to develop innovative footwear, including sensor-based insoles, aimed at preventing diabetic foot ulceration (DFU). The study focused on designing footwear that monitors key parameter such as pressure, temperature, and humidity. The first step involves identifying user requirements

through observational studies, followed by the creation and evaluation of semi-functional

prototypes. In the final step, a pre-clinical study assesses the prototypes' functionality. This human-centered design approach ensures the footwear meets user needs, reduces ulceration risk, and meets medical standards before proceeding to clinical trials. [7]

Raviglione et al. have developed a system that continuously monitors pressure on diabetic foot ulcers (DFUs) to improve the healing process by offloading pressure. A smart textile sensor integrated into an elastic band detects pressure and transmits data via Bluetooth to a connected device, providing real-time feedback through a phone app. The data have securely stored in the cloud, and the system's effectiveness was demonstrated through simulated patient testing. Although the system ensured the accuracy, it requires regular calibration and sensor replacement due to wear, and variability between sensors may affect precision in real-world clinical settings. [8]

Khandakar et al. have presented a novel insole design that integrates laser-induced graphene (LIG) temperature sensors to monitor foot sole temperature. With 20 LIG sensors per foot, the system offers comprehensive spatial coverage and high precision in detecting temperature variations. Real-time monitoring through thermogram images allows clinicians to identify early signs of DFUs, highlighting the system's potential in diabetic foot care. [9]

Abdul Razak et al. have compared platform-based and in-shoe foot pressure measurement systems, focusing on their applications, limitations, and sensor types, such as capacitive, resistive, and piezoresistive. They proposed Micro- Electromechanical Systems (MEMS) sensors for better accuracy, size, and efficiency. Wireless in-shoe systems are highlighted for real-time monitoring, especially in rehabilitation and sports science. However, in-shoe systems have limitations, such as lower spatial resolution and sensor slippage. The integration of MEMS increases costs and technical complexity, and power consumption issues persist in wireless systems. Sensor variability and environmental effects can impact data accuracy. [10]

Dragulinescu et al. have reviewed the advancements in smart socks and in-shoe systems for foot motion analysis, focused on sports and medical applications. The Pedar in-shoe system is emphasized for its ability to monitor foot pressure and gait, aiding in injury prevention, diabetic foot ulcer detection, and athletic training. The research work categorizes sensors used in these systems, such as capacitive, resistive, and piezoelectric types, discussing their roles in gait monitoring. Smart socks, which integrate sensors into textiles, are also highlighted for their medical and sports applications.

However, limitations include sensor placement inaccuracies, discomfort from textile integration, and issues with washing smart socks. In-shoe systems face challenges with sensor durability and high costs, while power consumption limits wireless systems' battery life. Environmental factors, such as indoor vs. outdoor conditions, also affect gait analysis accuracy. Despite

these drawbacks, the technologies show significant potential for improving foot pressure monitoring and injury prevention. [11]

Bagavathiappan et al. investigated the correlation between foot temperature and diabetic neuropathy using infrared thermal imaging. By analysing the relationship between foot temperature and vibration perception threshold (VPT), the study demonstrated that neuropathic patients exhibit significantly higher foot temperatures (32–35°C) than non-neuropathic patients (27–30°C). The findings suggested that real-time thermal imaging could serve as an early diagnostic tool for detecting neuropathy and preventing DFUs. [12]

Morey-Vargas and Smith investigate the use of infrared thermography as a non-invasive method to measure plantar foot temperatures in diabetic patients, correlating these measurements with neuropathy. The study, conducted on 112 type 2 diabetes patients in South India, found that diabetic neuropathic patients had higher mean foot temperatures (32–35°C) compared to non-neuropathic patients (27–30°C). Neuropathy was assessed using biothesiometry, with a significant correlation between higher vibration perception thresholds (VPT) and increased foot temperatures. However, there was no direct correlation between glycated haemoglobin (HbA1c) and foot temperatures, indicating that temperature variations are independent of overall blood sugar control. [13]

Rajendran et al. focused on the development of ultra-thin, flexible nanocomposite piezoresistive sensors for precise foot pressure measurement. These sensors, made from multi-walled carbon nanotubes (MWCNT) and polydimethylsiloxane (PDMS), offer high sensitivity (4.4 kΩ/kPa) and can measure pressures to 330 kPa with minimal hysteresis (1.7%), ensuring reliable repeated measurements. The sensors are integrated into a lightweight insole, providing valuable data on foot pressure distribution during activities like standing and walking.

The sensors maintain repeatability over 5000 cycles, demonstrating their potential for long-term monitoring of foot health and gait analysis. However, limitations include restricted sensor placement, the degradation of MWCNT dispersion over time, and sensitivity to ambient conditions like temperature and humidity. Additionally, the sensors may struggle with pressures exceeding 330 kPa, and they require an initial activation force to begin measuring, which could impact the accuracy of very low-pressure readings. [14]

Rami Basatneh discussed the potential of the Internet of Medical Things (IoMT) in improving care for diabetic foot ulcers (DFUs) by integrating smart devices and health sensors into home-care settings. IoMT enables real-time monitoring, remote care, and personalized interventions, reducing complications like infections and amputations. Innovations like voice-controlled technologies (e.g., Amazon Alexa) support diabetes management, helping patients monitor foot health, glucose levels, and medication adherence.

Wearable sensors and smart insoles are highlighted for tracking foot pressure and movement, preventing DFUs in high-risk patients. IoMT platforms offer predictive health

monitoring by detecting early signs of ulcers or infections. However, IoMT is still in its early stages, facing challenges such as data privacy concerns, patient

compliance, technical infrastructure needs, and limited clinical validation of voice-driven technologies. Issues with internet connectivity and device battery life further limit its widespread application. [15]

METHODS

PLANTAR PRESSURE MONITORING

Plantar pressure monitoring is crucial in preventing foot ulcers, particularly for individuals with diabetes or other conditions that impair sensation. Studies consistently highlight that high or prolonged pressure on specific areas of the foot can significantly elevate the risk of ulceration, leading to complications such as infections or amputations. To mitigate this risk, sensors embedded in insoles are used to continuously measure pressure at critical anatomical locations, including the toe, heel, and metatarsal heads. These sensors provide real-time data, enabling healthcare providers to identify excessive pressure points and make timely interventions. By analyzing pressure distribution patterns, clinicians can develop personalized offloading strategies, recommend appropriate footwear, and educate patients on the importance of regular foot inspections. Ultimately, effective plantar pressure monitoring fosters proactive management of foot health, improving overall outcomes and quality of life for at-risk individuals by significantly reducing the likelihood of ulcer development and associated complications.

TEMPERATURE AND HUMIDITY MONITORING

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TEMPERATURE AND HUMIDITY MONITORING

Monitoring temperature and humidity is vital for preventing foot ulcers in diabetic patients, as they are particularly susceptible to skin breakdown due to fluctuations in these parameters. Changes in foot temperature can indicate underlying issues such as inflammation or infection, making early detection essential. Temperature sensors are employed to continuously track foot temperature, allowing identification of deviations from the normal range. If a patient's foot temperature rises significantly, it may signal an impending ulcer or infection, prompting timely intervention. In addition to temperature monitoring, humidity sensors play a crucial role in assessing

moisture levels within footwear. Excessive moisture can weaken the skin barrier, increasing the risk of maceration and ulcer formation. By utilizing a combination of temperature and humidity sensors, healthcare providers can gain comprehensive insights into foot health. This data-driven approach enables personalized care strategies, ensuring that diabetic patients maintain optimal foot conditions, ultimately reducing the risk of ulceration and associated complications.

MACHINE LEARNING FOR PREDICTIVE ANALYSIS

Machine learning models, such as Gaussian Mixture Models (GMMs) and Multilayer Perceptrons (MLPs), are increasingly utilized for predictive analysis in assessing diabetic foot ulcer (DFU) risks. By analyzing plantar pressure distribution and related data, these models can identify patterns and anomalies that may indicate an increased risk of ulceration. They enable the reconstruction of detailed pressure maps, providing insights into how pressure is distributed across the foot. This advanced analysis enhances the accuracy of risk predictions, allowing healthcare providers to implement proactive interventions and personalized treatment plans, ultimately improving patient outcomes and reducing the incidence of DFUs.

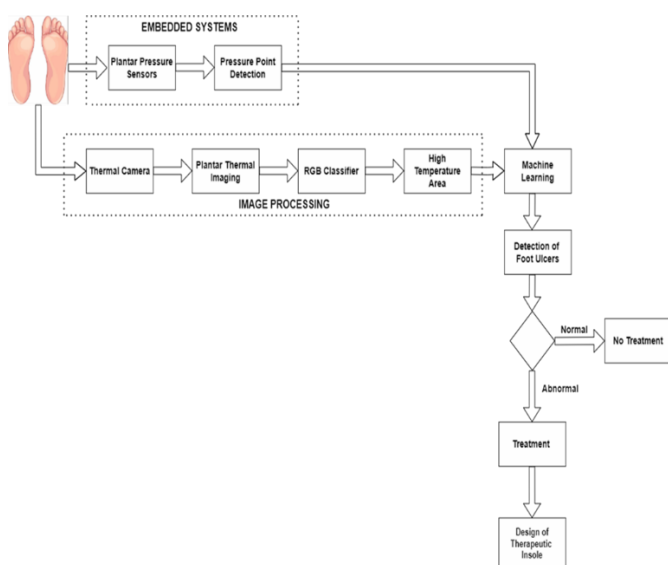


Fig. 1 A Frame Work for the proposed system

THERMAL IMAGING FOR EARLY DETECTION

Thermal imaging is emerging as a powerful tool for the early detection of temperature variations associated with neuropathy and foot ulcers in at-risk patients, particularly those with diabetes. By utilizing infrared thermal cameras or temperature sensors embedded in insoles, healthcare providers can continuously and non-invasively monitor foot temperature. This technology allows for the identification of localized heat patterns that may indicate inflammation or increased metabolic activity, which can precede ulcer formation. The ability to detect such temperature changes early enables timely intervention, reducing the risk of complications. Moreover, thermal imaging is advantageous as it does not require direct contact with the skin, making it a comfortable option for patients. This proactive approach enhances the management of foot health by facilitating prompt responses to developing issues, thereby potentially preventing serious conditions such as infections or amputations and improving overall patient outcomes.

REAL-TIME DATA TRANSMISSION

Real-time data transmission using Bluetooth and Bluetooth Low Energy (BLE) technology is integral to modern healthcare monitoring systems, particularly for patients at risk of diabetic foot ulcers. These technologies enable wireless communication between sensors embedded in insoles and mobile devices, facilitating seamless data transfer. The collected data, which includes critical metrics like pressure, temperature, and humidity, is processed in real-time, allowing both patients and clinicians to receive immediate insights into foot health. This timely transmission of information empowers healthcare providers to intervene promptly, implementing necessary changes to prevent ulcer development and ensuring better management of patient care through proactive monitoring.

IMPLEMENTATION

The development of a smart therapeutic insole system involves several key stages for the implementation:

SYSTEM DESIGN AND SENSOR INTEGRATION

The smart insole systems described in the related works incorporate a range of sensors, including force-sensitive resistors (FSRs), temperature sensors, and humidity sensors. The insole design typically focuses on monitoring critical anatomical locations on the foot, such as the toes, metatarsal heads, and heel, which are prone to increased pressure and ulcer formation in diabetic patients.

• PRESSURE SENSORS

FSRs are widely used to measure pressure at various points on the foot. These sensors detect abnormal pressure patterns, which can indicate areas at risk of ulceration.

• TEMPERATURE SENSORS

Temperature monitoring is crucial, as elevated foot temperature is often associated with inflammation and ulcer risk. Infrared and graphene-based temperature sensors are used to measure foot temperature in real-time, enabling early intervention.

• HUMIDITY SENSORS

Humidity is a less commonly monitored parameter, but it plays a significant role in the development of skin lesions and ulcers. By tracking foot humidity levels, smart insole systems can prevent excessive moisture build-up, which can lead to infections.

DATA ACQUISITION AND TRANSMISSION

The insole is equipped with advanced sensors that continuously collect real-time data on various foot conditions, such as pressure, temperature, and humidity. These data have to be wirelessly transmitted to a mobile application using Bluetooth or Bluetooth Low Energy (BLE) technology. BLE is particularly advantageous due to its low-power consumption, enabling prolonged use without frequent battery replacements. Once the data reaches the mobile app, it undergoes processing and visualization, allowing users to easily understand their foot health metrics. The app is designed to monitor these parameters closely, providing real-time alerts when any values exceed predefined thresholds. This immediate feedback is crucial for users, as it empowers them to take proactive measures, such as adjusting their footwear or seeking medical attention, to prevent complications like foot ulcers. Overall, this integrated approach

enhances user engagement in their health management and significantly improves the effectiveness of continuous foot monitoring for at-risk patients.

MACHINE LEARNING INTEGRATION

Machine learning algorithms, including Gaussian Mixture Models (GMMs), Multi-layer Perceptrons (MLP), and Principal Component Analysis (PCA), are employed to analyse data collected from foot sensors. These algorithms handle the complexity of high-dimensional data, enabling the generation of predictive models for assessing diabetic foot ulcer (DFU) risk. For instance, GMMs effectively model pressure distribution across the foot, identifying patterns associated with increased ulcer risk. Meanwhile, PCA is utilized to reduce data dimensionality, enhancing the efficiency and speed of the models. This integration of machine learning not only improves predictive accuracy but also facilitates timely interventions, ultimately aiding in the prevention of DFUs.

THERMAL IMAGING

Thermal imaging techniques, utilizing infrared cameras or embedded temperature sensors, are crucial components of the monitoring system designed for diabetic foot care. These technologies continuously track foot temperature variations, allowing for early detection of conditions such as neuropathy and potential ulceration by pinpointing abnormal hotspots on the foot. The integration of thermal imaging with machine learning algorithms significantly enhances the accuracy of ulcer prediction. By analyzing temperature data in conjunction with other metrics, machine learning models can identify patterns and risk factors that may lead to ulcer formation. This proactive approach not only facilitates timely interventions but also empowers patients to manage their foot health more effectively.

EVALUATION AND VALIDATION

The evaluation and validation process of the smart insole system is a critical step in establishing its effectiveness for monitoring foot health in diabetic patients. By rigorously assessing the system's accuracy, sensitivity, and specificity against established gold standard systems like Pedar, researchers can determine how reliably the smart insole measures pressure and detects potential issues. This comparative analysis is essential for demonstrating the system's clinical relevance and ensuring it meets high standards of performance. Additionally, gathering patient feedback provides valuable insights into usability and satisfaction, which are vital for user adoption and long-term success. Clinical trials further contextualize the system's performance in real-world scenarios, ensuring it can effectively monitor foot health outside of controlled environments. The use of statistical analysis methods, such as correlation coefficients and p-values, strengthens the findings by quantifying the reliability and significance of the results.



Fig 2: Thermal image of a 12 year old individual's foot without diabetes

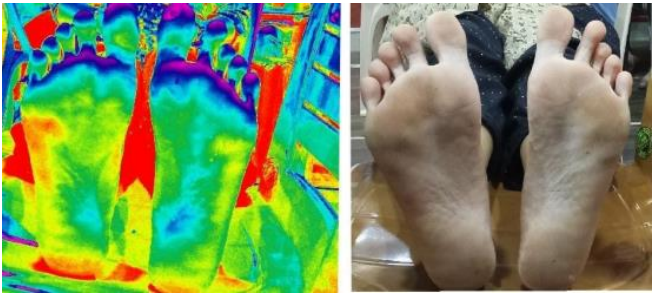


Fig 2: Thermal image of a 25 year old individual's foot without diabetes

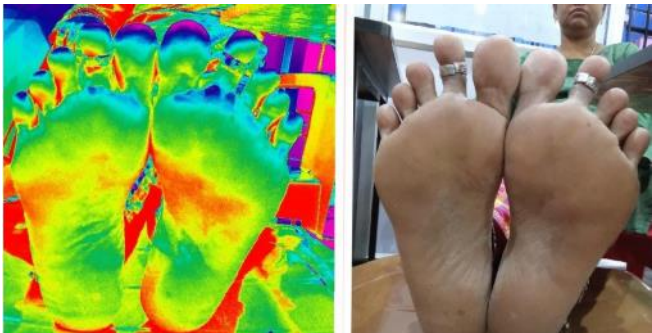
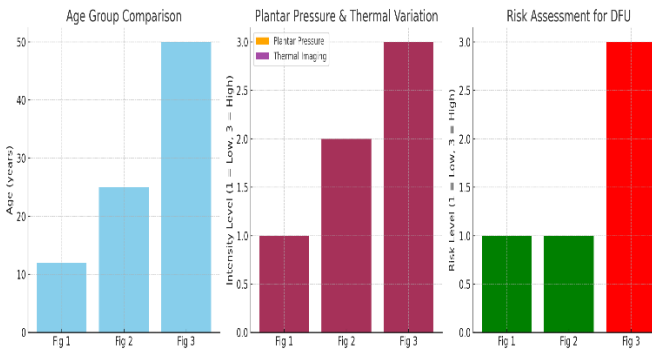


Fig 3: Thermal image of a 50 year old individual's foot with diabetes

TABLE 1: COMPARATIVE ANALYSIS OF FIGURES

Parameter	Fig 1	Fig 2	Fig 3
Age Group 1	12 years	25 years	50 years
Health Condition	Healthy, no diabetes, no DFU	Healthy, no diabetes, no DFU	Diabetic, existing foot ulcer
Plantar Pressure Analysis	Normal pressure distribution with slight variations	Concentration on heel and forefoot and Moderate, consistent pressures	High and uneven pressures - Significant elevation near ulcer site
Thermal Imaging	Even temperature distribution	Slightly elevated temperature at weight bearing spots and No abnormalities	Increased temperature near ulcer Inflammation and poor circulation visible
System Accuracy	No further intervention needed	No further intervention needed	Immediate medical attention recommended
Assessing Risks	Low risk for DFU	Low risk for DFU	High risk for DFU



CONCLUSION

The proposed frame work highlights a growing interest in smart insole systems to monitor plantar pressure and prevent diabetic foot ulcers (DFUs) using innovative sensor technologies and data analysis. Many of these systems, such as those using pressure sensors, temperature sensors, and Bluetooth for data transmission, aim to provide real-time monitoring to prevent DFUs. Machine learning methods, as proposed in some related research papers, significantly enhance the accuracy of these systems by reconstructing pressure maps or analyzing foot conditions. Moreover, thermal imaging techniques, as discussed in multiple studies, offer a promising non-invasive method for detecting foot ulcers early through temperature variations, often correlated with diabetic neuropathy. Our proposed frame work builds on these advancements by integrating a unique combination of pressure sensors, thermal cameras, and machine learning algorithms to provide a holistic solution for diagnosing diabetic foot conditions. Unlike other studies that focus on just plantar pressure or temperature monitoring, our frame work combines both modalities.

This dual approach enhances diagnostic accuracy by capturing both mechanical and thermal indicators of potential ulceration. By proposing this integrated system, we aim to address some of the gaps identified in existing studies, such as the need for simultaneous monitoring of multiple parameters (pressure, temperature, etc.) and a comprehensive diagnostic system that incorporates machine learning for better predictive capabilities. This proposed frame work has the potential to significantly reduce the occurrence of DFUs, providing a more effective solution for diabetic patients by improving early detection and targeted therapeutic intervention.

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